

ACADEMIC PROJECT PROPOSAL

StayPG

A MERN Stack Web Platform for Verified PG/Hostel Discovery and Booking for Students in Patiala

*An Airbnb-style marketplace connecting PG/hostel owners with
verified students, built on MongoDB, Express.js, React.js and Node.js*

Field	Details
Project Title	StayPG – PG/Hostel Discovery & Booking Platform
Domain	Full Stack Web Development (MERN)
Target City	Patiala, Punjab
Duration	12 Weeks (One Semester)
Team Size	As per course requirement (Solo / 2–4 members)
Submitted To	[Course Instructor / Department Name]

1 Introduction / Problem Statement

Every academic year, thousands of students relocate to Patiala to attend its colleges and universities, and nearly all of them need to find a Paying Guest (PG) accommodation or hostel within their first few weeks. At present this search is handled almost entirely offline: students rely on word-of-mouth, local property agents, or scattered posts on WhatsApp and Facebook groups. There is no single, trustworthy platform where a student can compare verified listings, view real photographs, check genuine reviews from previous tenants, and book a room with confidence.

This creates two connected problems. Students frequently arrive in an unfamiliar city and are forced to make a rushed, high-stakes housing decision based on incomplete or unverifiable information, which exposes them to overpriced rent, misrepresented rooms, and even fraudulent listings. At the same time, genuine PG and hostel owners have no dedicated digital channel to showcase their property, manage bookings, or build a reputation through reviews – they remain dependent on brokers and informal referrals, which limits their occupancy and revenue.

This project proposes **StayPG**, a full-stack MERN (MongoDB, Express.js, React.js, Node.js) web application modelled on the Airbnb marketplace concept but purpose-built for the student PG/hostel segment of Patiala. The platform will allow verified owners to list their property with photographs, pricing, and amenities, and will allow students to search, filter, locate on a map, message the owner, read and post reviews, and book a stay – all secured by proper authentication and role-based authorization.

2 Objectives

Primary Objective: To design and develop a secure, full-stack MERN web application that connects PG/hostel owners in Patiala with students through verified property listings, search and booking, in-app communication, and a review system.

Specific Objectives:

- To build an owner dashboard for listing PG/hostel properties with multiple images, rent, room type, amenities and location.
- To implement student-facing search and filter (price range, distance from college, gender preference, sharing type, amenities).
- To integrate map-based property location using a geolocation/maps API so students can view distance from their institute.
- To implement a booking and request-management workflow with real-time availability status for each property.
- To build a review and rating system so students can rate and review a PG after a verified stay/booking.
- To implement an owner-verification workflow (ID/document check by admin) so listings display a “Verified Owner” badge.
- To enable secure, real-time chat between a student and an owner for queries prior to booking.
- To implement JWT-based authentication and role-based authorization for three roles:

Student, Owner, and Admin.

3 Methodology

As this is an engineering / software development project rather than a research study, the methodology follows a system architecture and iterative build-and-test approach, moving from environment setup, to module development, to integration, to validation.

3.1 System Architecture & Tech Stack

Layer	Technology	Purpose
Frontend	React.js, React Router, Redux Toolkit / Context API, Tailwind CSS / Material UI	Single-page application, state management, responsive UI
Backend	Node.js, Express.js (REST API)	Business logic, routing, middleware, API endpoints
Database	MongoDB with Mongoose ODM	Storing users, properties, bookings, reviews, chats
Authentication	JWT (JSON Web Tokens), bcrypt.js, HTTP-only cookies	Login, signup, password hashing, session/token handling
Authorization	Custom Express middleware (role-based access control)	Restricting routes by role: Student / Owner / Admin
Image Storage	Multer + Cloudinary (or AWS S3)	Uploading and serving property images
Maps & Location	Google Maps JavaScript API / Geocoding API	Pin property location, distance from colleges, “near me” search
Real-time Chat	Socket.io	Live messaging between student and owner
Deployment	Vercel/Netlify (frontend), Render/Railway (backend), MongoDB Atlas (DB)	Cloud hosting for demonstration and testing
Testing	Postman, Jest, React Testing Library	API testing, unit testing, UI testing

3.2 Module Development

a) User & Role Management Module

- Registration/login for three roles – Student, Owner, Admin – with separate dashboards.
- Profile management: student verification details, owner business/ID details.

b) Property Listing Module (Owner side)

- CRUD operations for listings: title, description, rent, deposit, sharing type, amenities.
- Multi-image upload with preview, stored via Cloudinary and referenced in MongoDB.

- Availability toggle (vacant/occupied) and room-count management.

c) Search, Filter & Locate Module (Student side)

- Filter by price, gender preference, distance, amenities and rating.
- Interactive map view showing pins for nearby PGs relative to a selected college.

d) Booking Module

- Student sends a booking/visit request; owner accepts, rejects, or proposes an alternate date.
- Booking history and status tracking (Pending, Confirmed, Cancelled, Completed).

e) Review & Verification Module

- Students can rate and review a property only after a completed booking (prevents fake reviews).
- Admin panel to verify owner documents and approve/reject listings before they go live.

f) Chat Module

- Real-time one-to-one chat between student and owner using Socket.io, with chat history stored in MongoDB.

3.3 Execution & Integration – Authentication and Authorization Flow

- Signup/login issues a short-lived JWT access token plus a refresh token stored in an HTTP-only cookie.
- Passwords are hashed with bcrypt before being stored; raw passwords are never persisted.
- Protected Express routes use an auth middleware that verifies the JWT on every request.
- A role-based authorization middleware restricts endpoints – e.g. only an Owner can create/edit a listing, only an Admin can verify owners, only the Student who made a booking can post a review for it.
- Sensitive actions (delete listing, verify owner) require re-validation of role and ownership of the resource, not just a valid token.

3.4 Testing & Validation

- **Unit Testing:** individual functions/components (e.g., booking-status logic, rating calculation) tested with Jest.
- **API Testing:** all REST endpoints tested in Postman, including negative tests for unauthorized access.
- **Integration Testing:** end-to-end flow from signup → listing creation → search → chat → booking → review.
- **User Acceptance Testing (UAT):** a small sample of real students and PG owners in Patiala trial the platform and give feedback.
- **Security Testing:** checks for broken access control, JWT tampering, and unauthorized role escalation.

3.5 Constraints & Safety

- **Data Privacy:** student contact details and owner documents are only visible to authorized, verified parties.
- **Listing Trust:** unverified owners cannot publish a live listing, reducing the risk of fraudulent posts.
- **Scalability:** image storage and chat are offloaded to Cloudinary/Socket.io so the core API remains lightweight.
- **Map/API limits:** free-tier Google Maps API usage will be monitored to remain within the free monthly quota.

4 Timeline

The project will be executed over a 12-week semester schedule, with an internal buffer built into the final week.

Week	Phase	Task	Deliverable
1–2	Planning	Requirement gathering, DB schema design, wireframes/UI mockups	SRS document + Figma wireframes
3–4	Backend Setup	Node/Express project setup, MongoDB schemas, JWT auth & role middleware	Working auth APIs (signup/login/roles)
5–6	Core Backend	Property CRUD APIs, image upload (Multer/Cloudinary), search/filter APIs	Property & search REST API
7–8	Frontend Build	React UI: landing page, listing page, property detail, owner dashboard	Functional frontend connected to back-end
9	Booking & Reviews	Booking workflow, review/rating module, admin verification panel	End-to-end booking & review flow
10	Chat & Maps	Socket.io real-time chat, Google Maps location integration	Live chat + map-based locate feature
11	Testing	Unit, API, integration and security testing; bug fixing	Test report + stable build
12	Deployment & Buffer	Cloud deployment, documentation, demo prep, contingency buffer	Deployed app + final report/presentation

5 Resources / Budget

Human Resources: If completed as a group, roles will be divided as Frontend Developer, Backend Developer, and Database/DevOps lead; each member also contributes to testing and documentation. If completed individually, all roles are handled sequentially by the sole developer.

Material / Technical Resources:

- A personal laptop with Node.js, MongoDB Compass, VS Code, and Git installed.
- MongoDB Atlas free-tier cluster for the database.
- Cloudinary free tier for image storage.
- Google Maps Platform free monthly credit for geolocation/maps.
- GitHub for version control and Postman for API testing.

Budget:

Item	Estimated Cost	Notes
MongoDB Atlas / Clouinary / Render	Rs. 0 (Free tier)	Sufficient for a semester-scale demo project
Google Maps API	Rs. 0 (Free monthly credit)	Usage capped to stay within free quota
Domain name (optional)	≈ Rs. 800/year	Only if a custom domain is desired for the demo
Total	Rs. 0 – Rs. 800	No external funding required; all core resources are available on free tiers

6 Elevator Pitch Summary

- **The Gap:** Currently, students moving to Patiala have no verified, centralised platform to find trustworthy PG/hostel accommodation.
- **The Solution:** This project will build StayPG, a MERN-stack marketplace where verified owners list properties with images and students can search, locate, chat, book and review – all behind secure authentication and role-based authorization.
- **The Impact:** This will give Patiala's student community a safer, more transparent way to find housing, while giving genuine PG owners a proper digital channel to reach verified tenants.